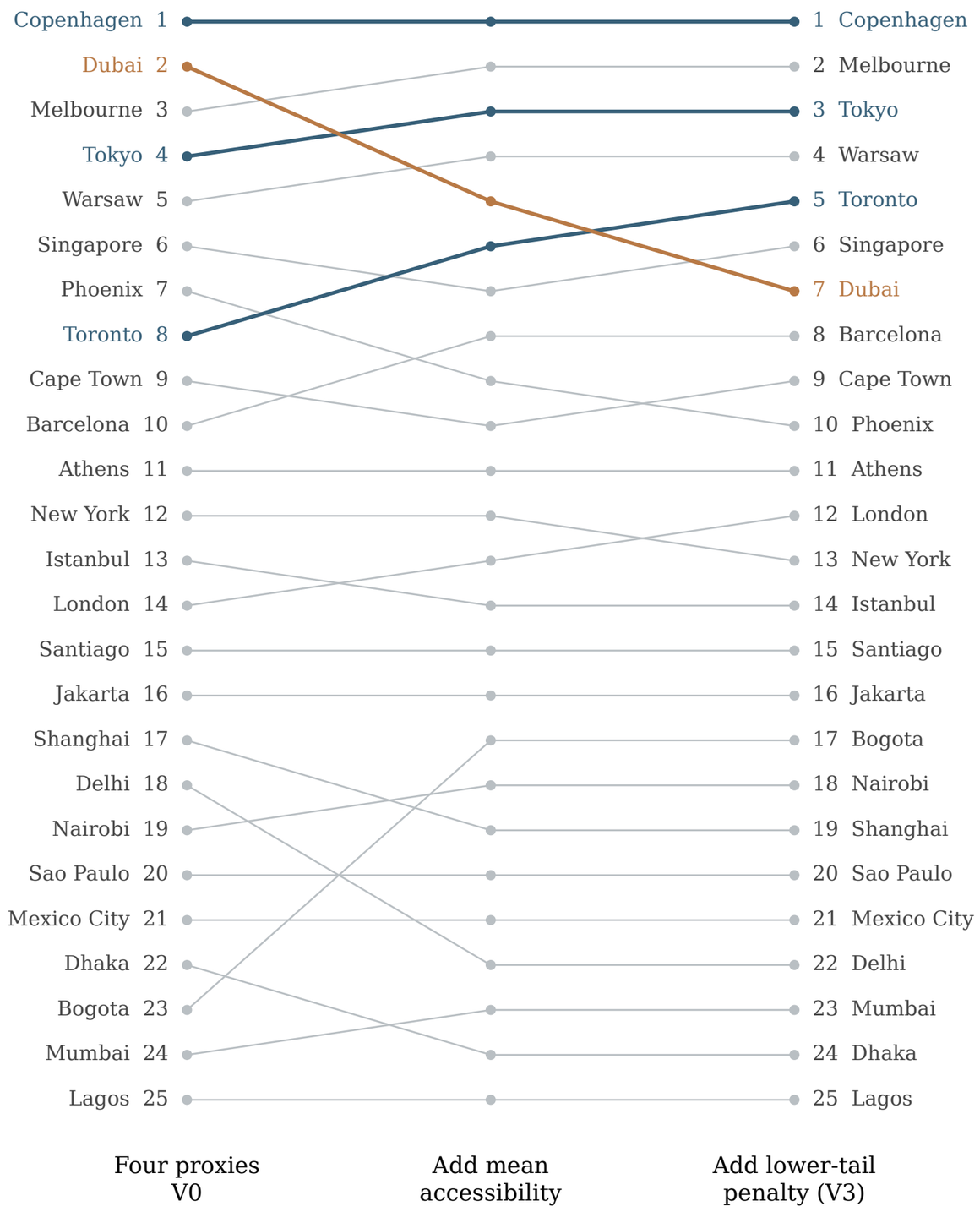
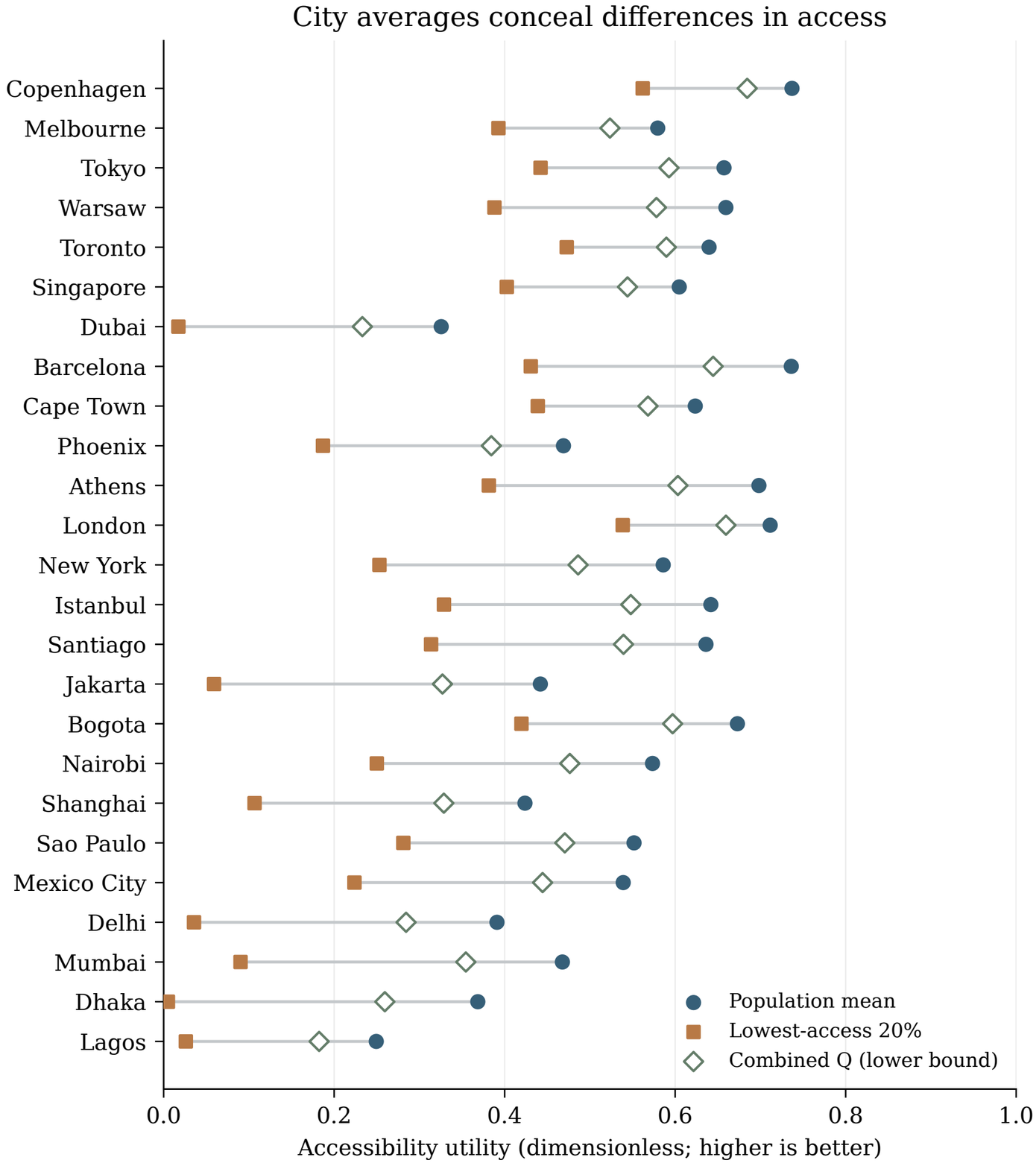
## Rank changes after adding learning-facility accessibility




25-city archived analysis. Ranking changes reflect a changed evaluation model, not measured improvement.




Model-derived resident learning-facility proxy; not school quality. Walking; tau = 30 min; tail weight = 0.30.

Before: access + education unknown

After: access observed; education unknown

Copenhagen
Melbourne
Tokyo
Warsaw
Toronto
Singapore
Dubai
Barcelona
Cape Town
Phoenix
Athens
London
New York
Istanbul
Santiago
Jakarta
Bogota
Nairobi
Shanghai
Sao Paulo
Mexico City
Delhi
Mumbai
Dhaka
Lagos

20 30 40 50 60 70 80

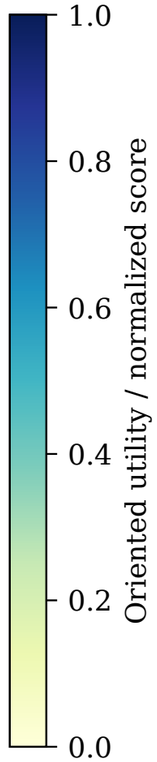
Feasible score interval

Possible champion

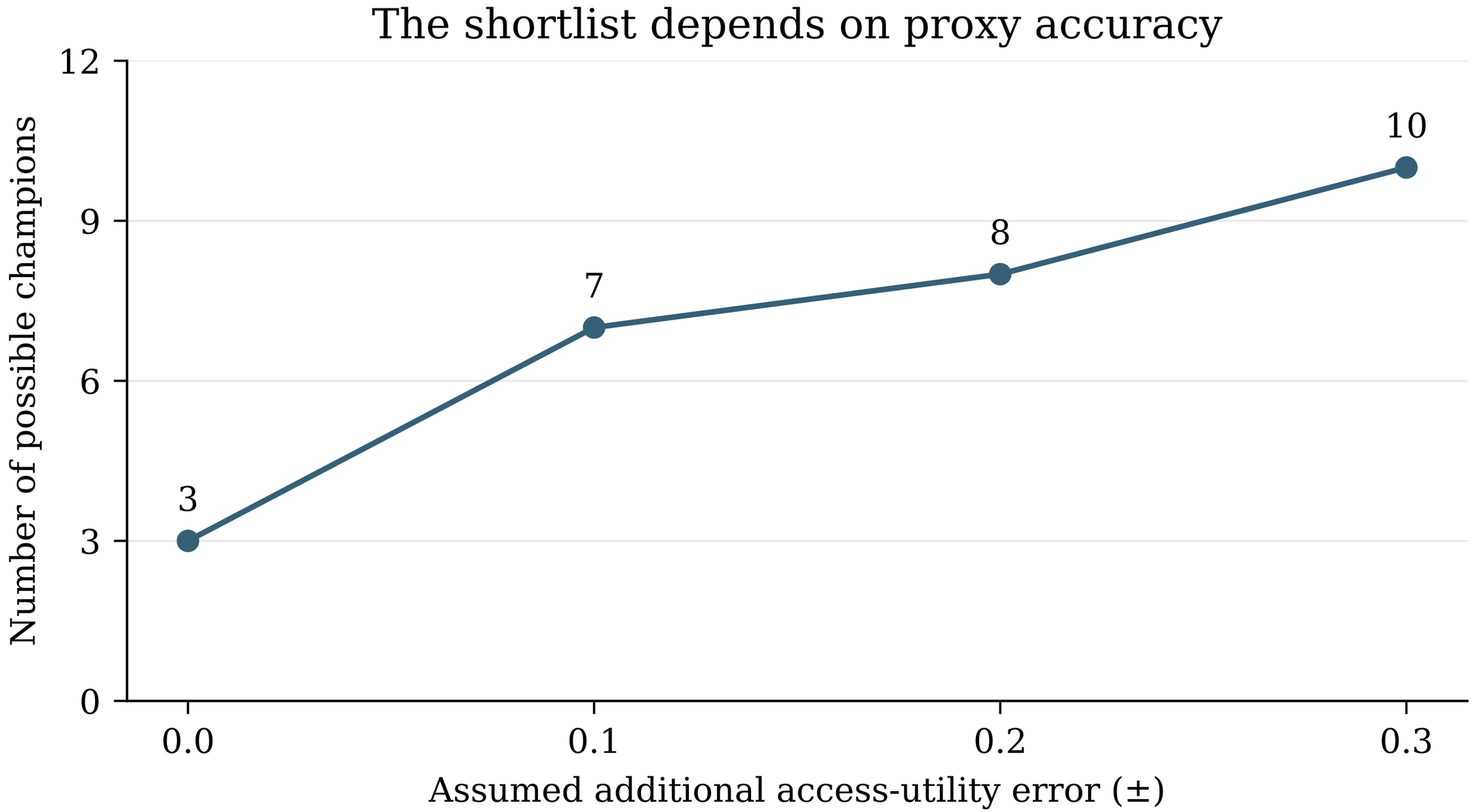
Same model and weights in both panels: four proxies 80%, access 10%, unknown education 10%. Not confidence intervals.

## Five proxy indicators across the 25-city sample

| | Safety | Cleanliness | Housing affordability | Commute | Learning access (Q) |
|---|---|---|---|---|---|
| Copenhagen | 0.74 | 0.79 | 0.79 | 0.69 | 0.68 |
| Melbourne | 0.56 | 0.73 | 0.81 | 0.54 | 0.52 |
| Tokyo | 0.75 | 0.57 | 0.60 | 0.52 | 0.59 |
| Warsaw | 0.74 | 0.40 | 0.66 | 0.61 | 0.58 |
| Toronto | 0.57 | 0.62 | 0.69 | 0.50 | 0.59 |
| Singapore | 0.77 | 0.67 | 0.42 | 0.55 | 0.54 |
| Dubai | 0.84 | 0.51 | 0.83 | 0.60 | 0.23 |
| Barcelona | 0.48 | 0.36 | 0.69 | 0.66 | 0.64 |
| Cape Town | 0.26 | 0.61 | 0.87 | 0.53 | 0.57 |
| Phoenix | 0.47 | 0.42 | 0.90 | 0.64 | 0.38 |
| Athens | 0.45 | 0.45 | 0.63 | 0.58 | 0.60 |
| London | 0.45 | 0.42 | 0.53 | 0.50 | 0.66 |
| New York | 0.49 | 0.42 | 0.65 | 0.52 | 0.49 |
| Istanbul | 0.52 | 0.32 | 0.64 | 0.43 | 0.55 |
| Santiago | 0.36 | 0.29 | 0.55 | 0.59 | 0.54 |
| Jakarta | 0.47 | 0.15 | 0.71 | 0.41 | 0.33 |
| Bogota | 0.34 | 0.30 | 0.45 | 0.42 | 0.60 |
| Nairobi | 0.40 | 0.20 | 0.57 | 0.41 | 0.48 |
| Shanghai | 0.72 | 0.29 | 0.15 | 0.49 | 0.33 |
| Sao Paulo | 0.30 | 0.21 | 0.55 | 0.44 | 0.47 |
| Mexico City | 0.33 | 0.19 | 0.56 | 0.41 | 0.44 |
| Delhi | 0.41 | 0.10 | 0.72 | 0.36 | 0.28 |
| Mumbai | 0.56 | 0.17 | 0.29 | 0.39 | 0.35 |
| Dhaka | 0.38 | 0.06 | 0.67 | 0.32 | 0.26 |
| Lagos | 0.32 | 0.11 | 0.00 | 0.22 | 0.18 |

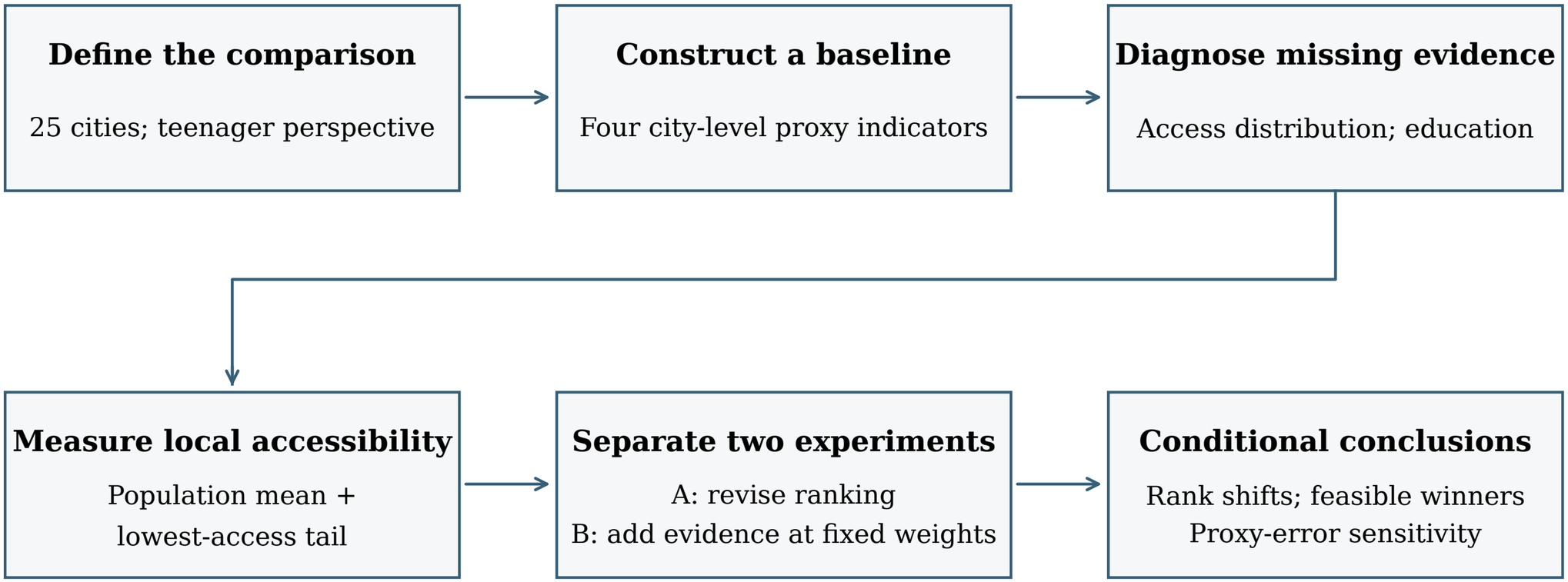



Columns are proxy measures, not direct student outcomes. Higher is better. City order follows the V3 ranking.




Stress assumptions, not estimated measurement errors. Education remains unknown; this is not a probability curve.

An evidence-led revision of the city evaluation model




Research workflow proposal. Education quality, eligibility and teenager-specific travel remain unresolved.

## 1. Define the comparison and its evidence

25 cities
Common sample

Safety
Cleanliness
Housing burden
Adult commute

Proxy matrix
$z_{ij}$

*Age-specific education evidence remains missing.*

## 2. Establish the city-level baseline

Orient and normalize benefit / cost indicators

Equal baseline weights
$w_j = 1/4$

Fixed-ideal TOPSIS
$C_i = D_i^- / (D_i^- + D_i^+)$

V0 ranking

*Baseline winner: Copenhagen*

## 3. Measure access beyond the city average

Resident grids
$(p_{ir}, t_{ir})$

Travel time to learning POIs
$u_{ir} = \exp(-t_{ir}/30)$

Population mean $\mu_i$

Lowest 20% $L_i$

$$Q_i = 0.7\mu_i + 0.3L_i$$

## 4. Separate model revision from information gain

A. Change the model
Five proxies, 20% each

16 / 25 ranks change

Measurement + lower tail

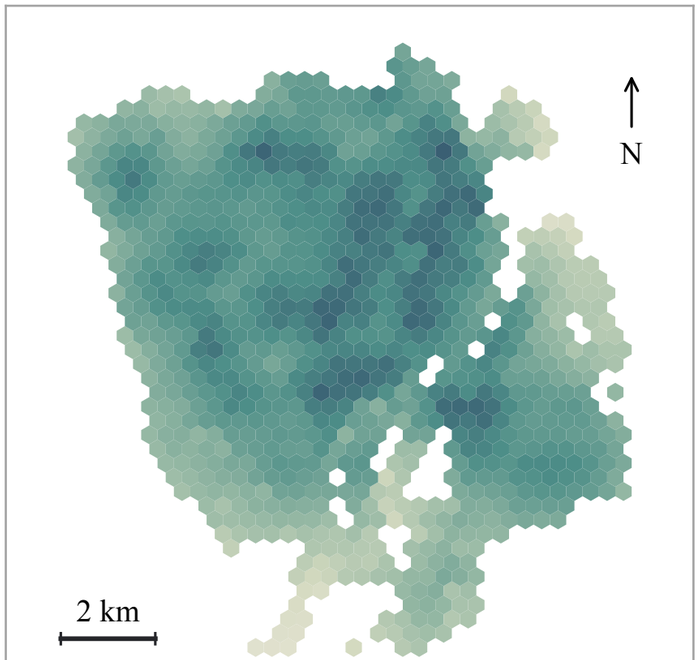
B. Hold weights fixed
80% + 10% + 10%

Possible winners: 12 → 3

Education stays unknown

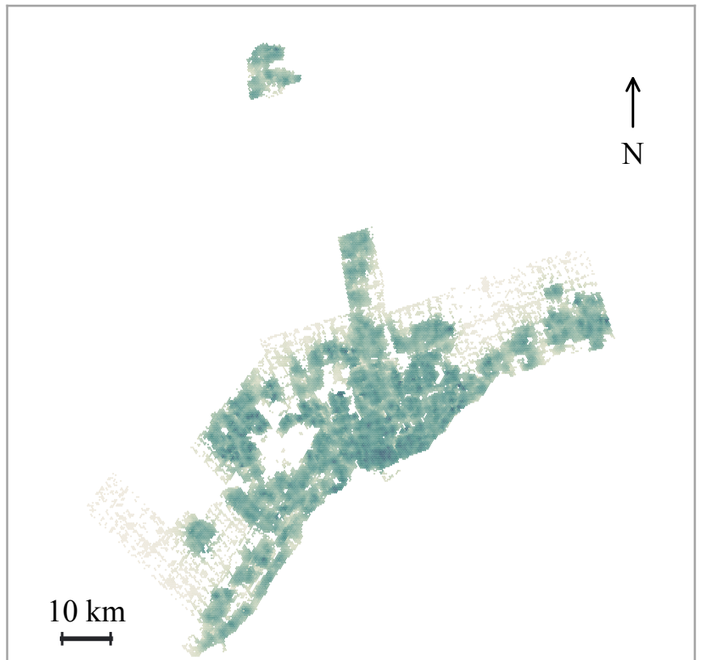
*Check parameter choices and proxy-error assumptions.*

(a) Copenhagen



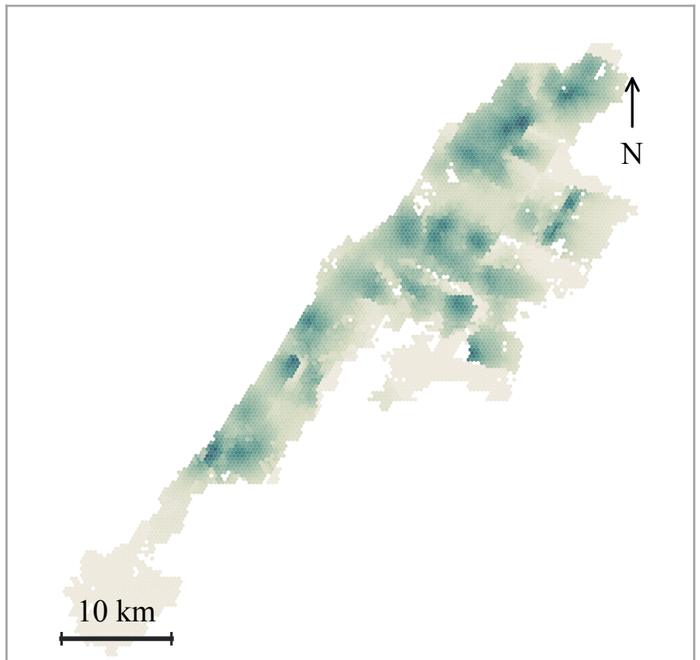

$\mu = 0.737$ $L = 0.562$

(b) Toronto

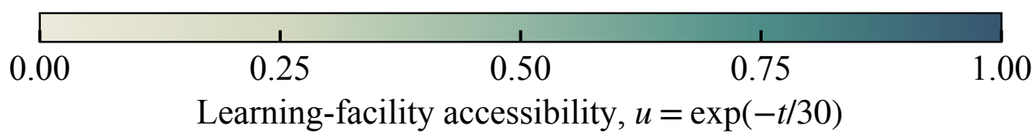



$\mu = 0.640$ $L = 0.473$

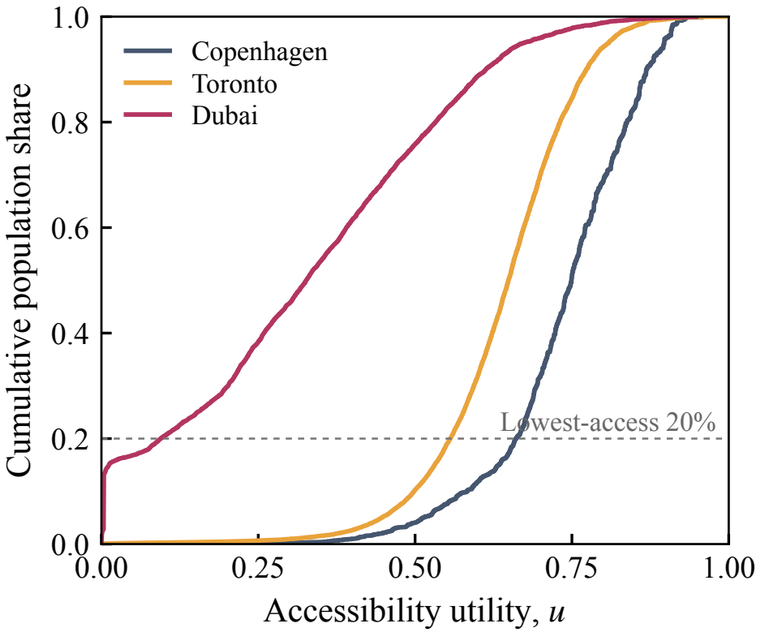
(c) Dubai



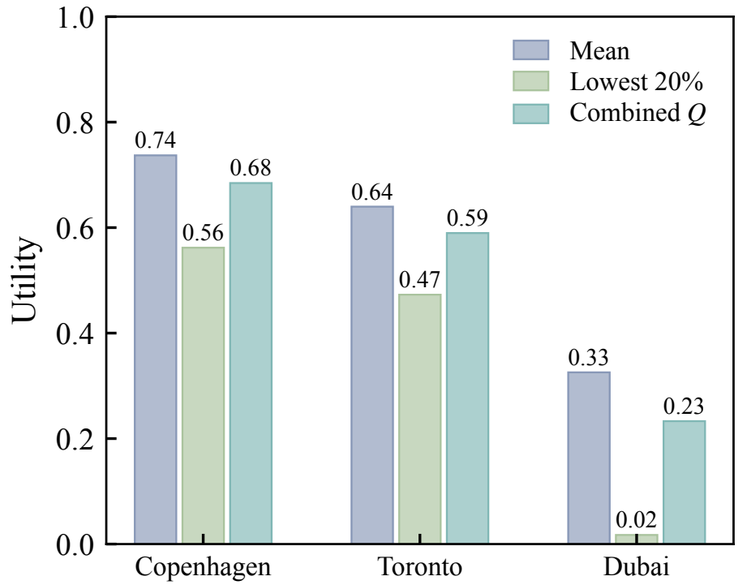

$\mu = 0.326$ $L = 0.017$



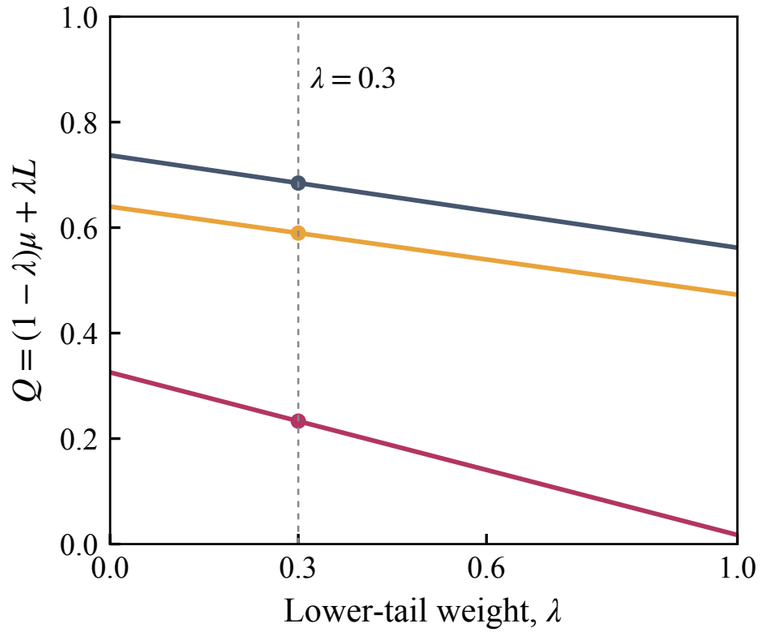
(a) Population-weighted distribution



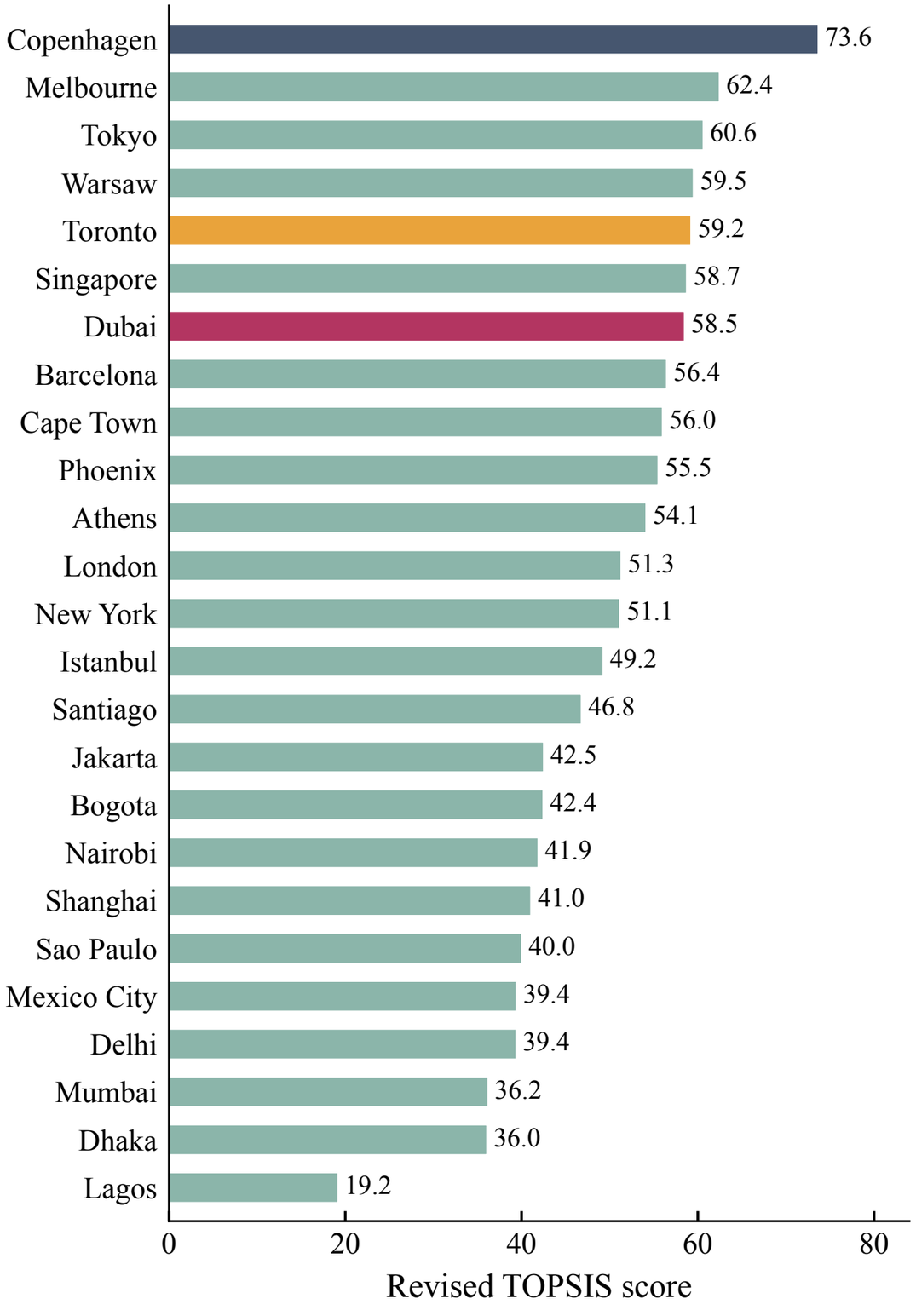

(b) Mean and lower-tail aggregation



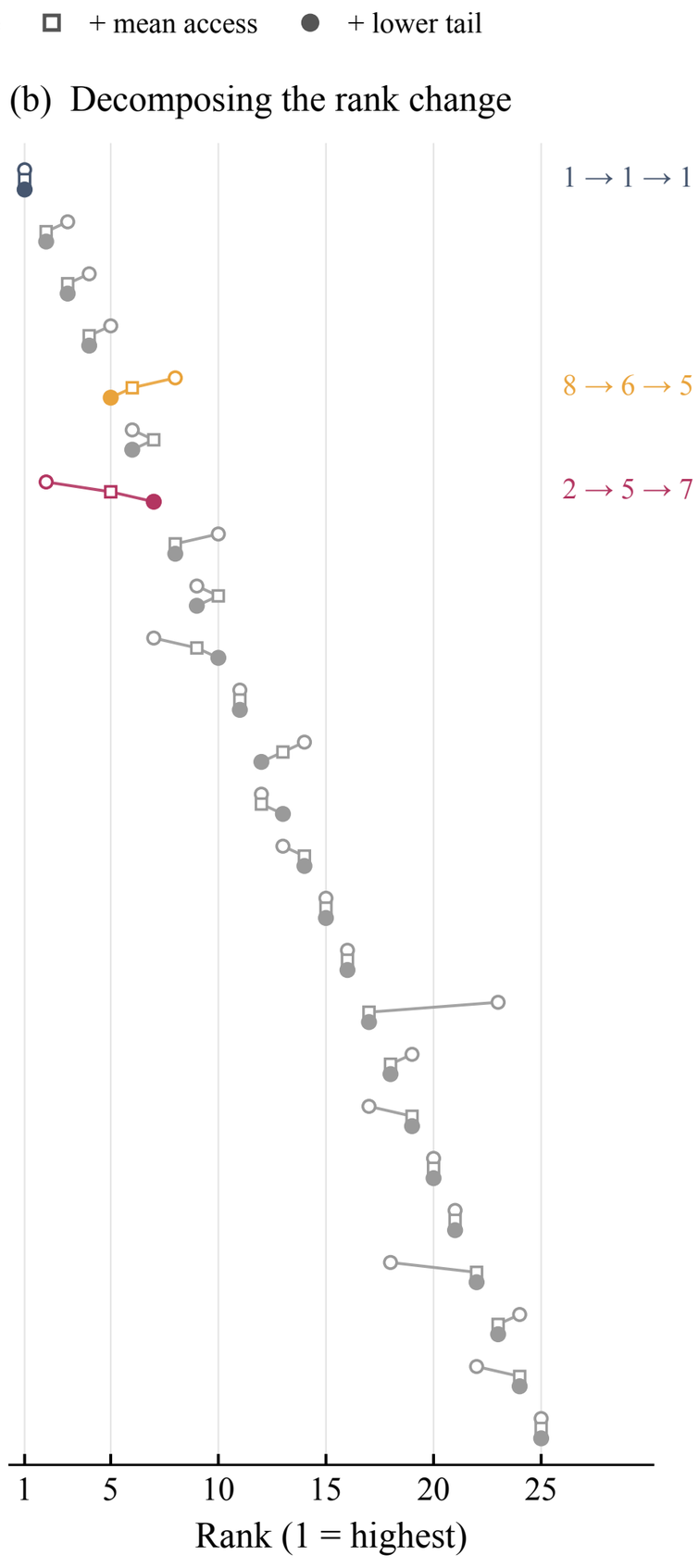

(c) Effect of the equity preference



(a) Five-proxy ranking (all 25 cities)



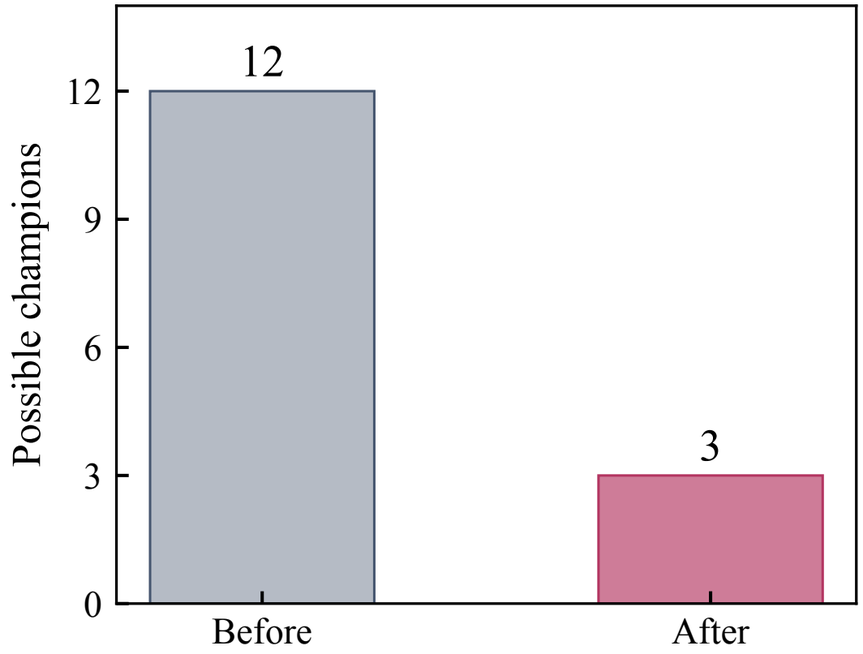

(b) Decomposing the rank change



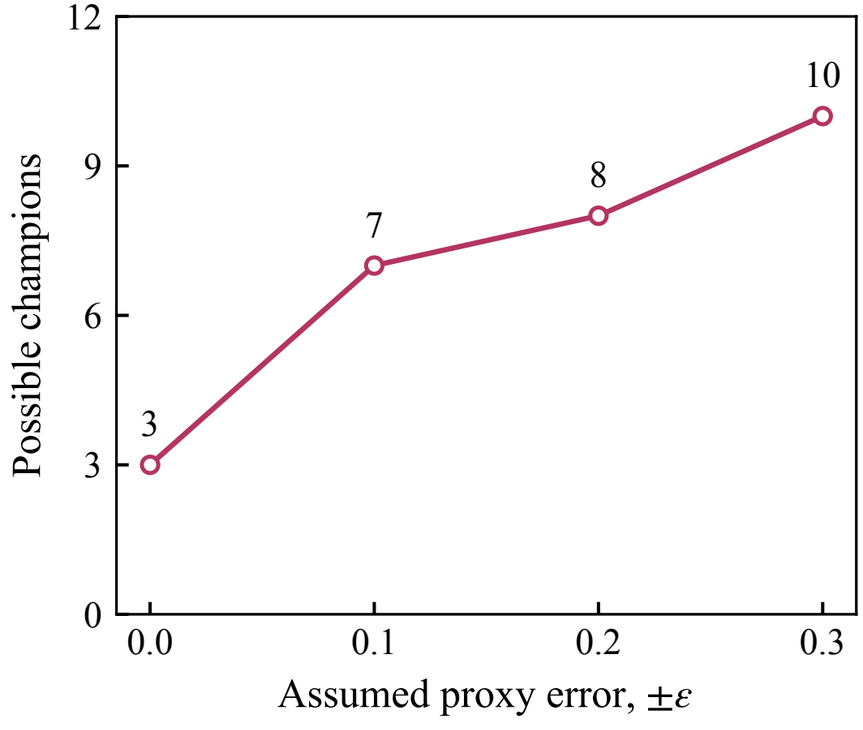
(a) Observed access narrows the score bounds




(b) Same model, additional evidence




(c) Sensitivity to proxy error



| | Copenhagen | Melbourne | Tokyo | Warsaw | Toronto | Singapore | Dubai | Barcelona | Cape Town | Phoenix | Athens | London | New York | Istanbul | Santiago | Jakarta | Bogota | Nairobi | Shanghai | Sao Paulo | Mexico City | Delhi | Mumbai | Dhaka | Lagos |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Safety | 0.74 | 0.56 | 0.75 | 0.74 | 0.57 | 0.77 | 0.84 | 0.48 | 0.26 | 0.47 | 0.45 | 0.45 | 0.49 | 0.52 | 0.36 | 0.47 | 0.34 | 0.40 | 0.72 | 0.30 | 0.33 | 0.41 | 0.56 | 0.38 | 0.32 |
| Cleanliness | 0.79 | 0.73 | 0.57 | 0.40 | 0.62 | 0.67 | 0.51 | 0.36 | 0.61 | 0.42 | 0.45 | 0.42 | 0.42 | 0.32 | 0.29 | 0.15 | 0.30 | 0.20 | 0.29 | 0.21 | 0.19 | 0.10 | 0.17 | 0.06 | 0.11 |
| Housing affordability | 0.79 | 0.81 | 0.60 | 0.66 | 0.69 | 0.42 | 0.83 | 0.69 | 0.87 | 0.90 | 0.63 | 0.53 | 0.65 | 0.64 | 0.55 | 0.71 | 0.45 | 0.57 | 0.15 | 0.55 | 0.56 | 0.72 | 0.29 | 0.67 | 0.00 |
| Commute | 0.69 | 0.54 | 0.52 | 0.61 | 0.50 | 0.55 | 0.60 | 0.66 | 0.53 | 0.64 | 0.58 | 0.50 | 0.52 | 0.43 | 0.59 | 0.41 | 0.42 | 0.41 | 0.49 | 0.44 | 0.41 | 0.36 | 0.39 | 0.32 | 0.22 |
| Learning access | 0.68 | 0.52 | 0.59 | 0.58 | 0.59 | 0.54 | 0.23 | 0.64 | 0.57 | 0.38 | 0.60 | 0.66 | 0.49 | 0.55 | 0.54 | 0.33 | 0.60 | 0.48 | 0.33 | 0.47 | 0.44 | 0.28 | 0.35 | 0.26 | 0.18 |

0.00 0.25 0.50 0.75 1.00

Oriented proxy score / utility (higher is better)